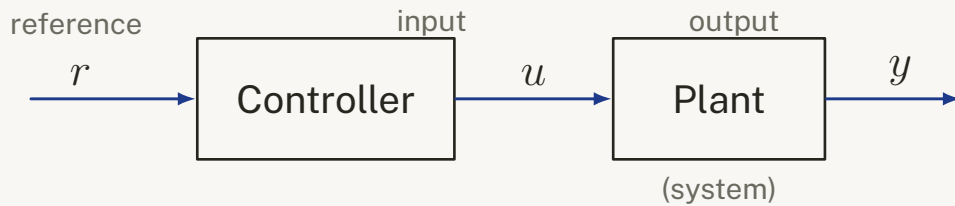




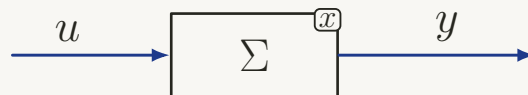
Linearization

RECAP

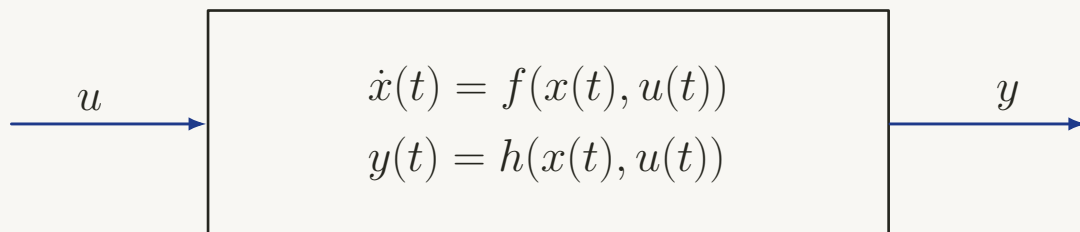
General structure of our problem:



To solve our problem, we need to understand our system:



Which is really just...



... and we can find $f(x(t), u(t))$ and $h(x(t), u(t))$ through modeling.

1 | System classification

We can classify our system in four different ways.

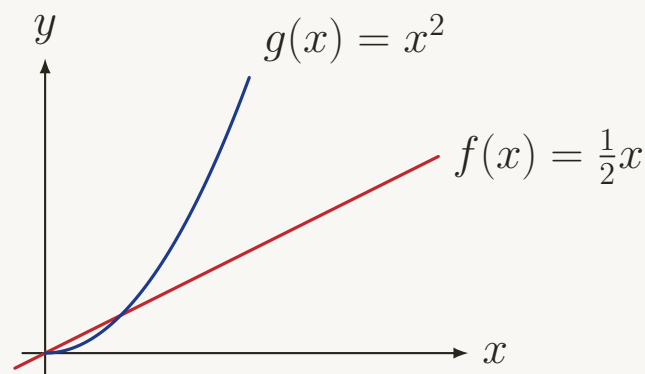
i) Linearity

DEF a system is **linear** if it is...

... **Additive:** $\Sigma(u_1 + u_2) = \Sigma(u_1) + \Sigma(u_2)$

... **Homogeneous:** $\Sigma(\alpha u) = \alpha \Sigma(u)$

INTUITION



... why do we care? **Superposition!**

$$y = \Sigma(u)$$

$$\rightarrow u = \alpha u_1 + \alpha u_2$$

$$\Rightarrow y = \Sigma(u) = \Sigma(\alpha u_1 + \alpha u_2) = \alpha \Sigma(u_1) + \alpha \Sigma(u_2)$$

$$:= y_1 + y_2$$

ii) Causality

DEF a system is **causal** if, for all t , $y(t)$ only depends on $(-\infty, t]$. It is **strictly causal** if $y(t)$ only depends on $(-\infty, t)$.

EX.

Causal

$$y(t) = 3u(t)$$

$$y(t) = t \cdot \cos(t - 3)$$

Non-causal

$$y(t) = u(2t)$$

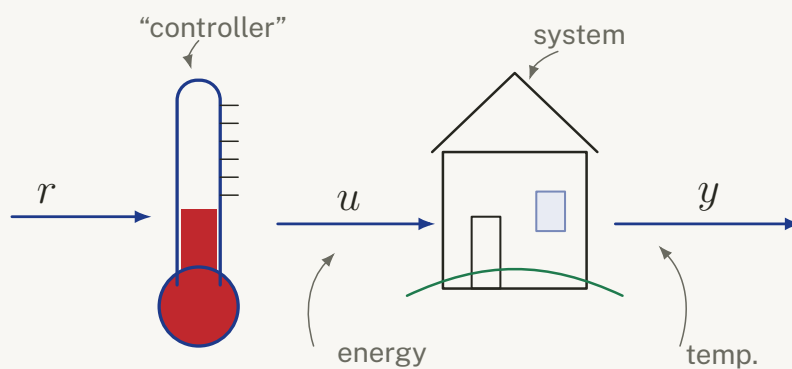
$$y(t) = \int_{-\infty}^{t+1} u(\tau) d\tau$$

$$y(t) = (t + 2) \cos(u(t))$$

INTUITION All practically realizable systems are causal! (impossible to implement a non-causal system in “the real world”)

A thermostat, for instance, reacts only to temperatures it has already measured.

EX.



iii) Memoryless (static) vs. dynamic systems

DEF a system is **memoryless (static)** if $y(t)$ only depends on $u(t)$: the output at the present time only depends on the input at the present time.

EX.

Static

$$y(t) = 3u(t)$$

$$y(t) = 2^{-(t+1)}u(t)$$

$$y(t) = \sqrt{\sin(u^2(t))}$$

Dynamic

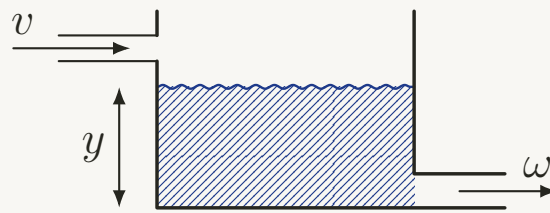
$$y(t) = \int_{-\infty}^t u(\tau) d\tau$$

$$y(t) = \dot{u}(t)$$

$$y(t) = u(2t)$$

Rule of thumb: an integral or a derivative in the equation always means the system is dynamic.

INTUITION the water tank



v inflow · w outflow · y water level

$$u = v - w$$

$$y(t) = \int_{-\infty}^t u(\tau) d\tau$$

$\Rightarrow$ acts like an **integral** — so it's **dynamic**!

iv) Time-variance

DEF a system is **time-invariant** if the same input produces the same output at any time.

EX.

Time-invariant

$$y(t) = \cos(u(t))$$

$$y(t) = 8u(t)$$

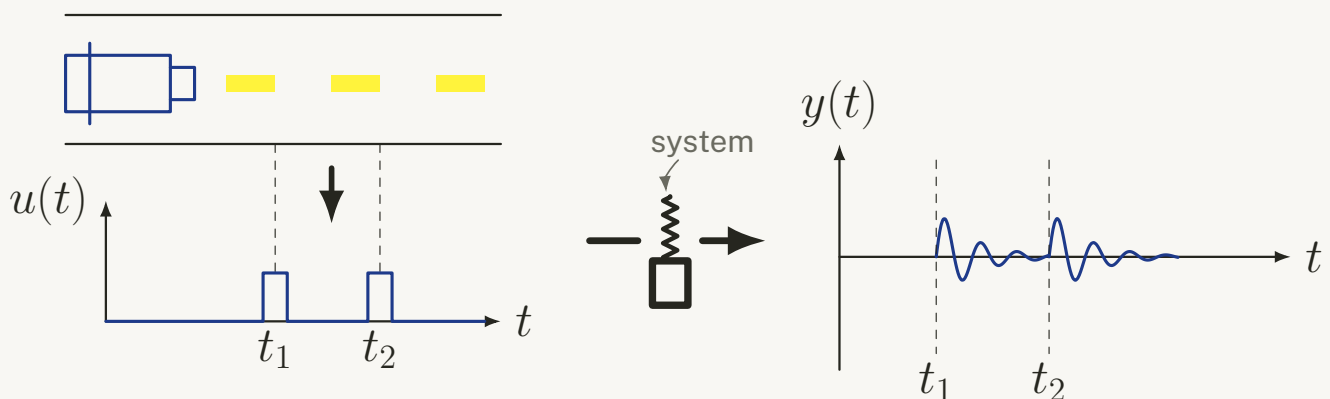
Time-variant

$$y(t) = 3t \cos(t)$$

$$y(t) = \cos(t) \cdot u(t)$$

INTUITION (car-damping system):

Hitting the same bump at different times should produce the same oscillation in the output — just shifted in time.



QUIZ

Q1) $y(t) = \sqrt{t - \cos^2(t)} \cdot u(t^2)$

Statement	True	False
The system is linear	×	
The system is static		×
The system is time-invariant		×

Q2) $u(t) \in \mathbb{R}^2, y(t) \in \mathbb{R}$

$x(t) = u_1(t - T)$, mit der Konstante $T = 1$ s

$y(t) = -x^2(t) + 2 \cdot u_2(t)$

Behauptung	wahr	falsch
Das System ist linear		×
Das System ist zeit-invariant	×	
Das System ist dynamisch	×	
Das System ist SISO		×

Q3) $x_1(t) = \int_{-\infty}^t u(\tau - 3) d\tau, x_2(t) = u(t), y(t) = x_1(t) + x_2(t)$

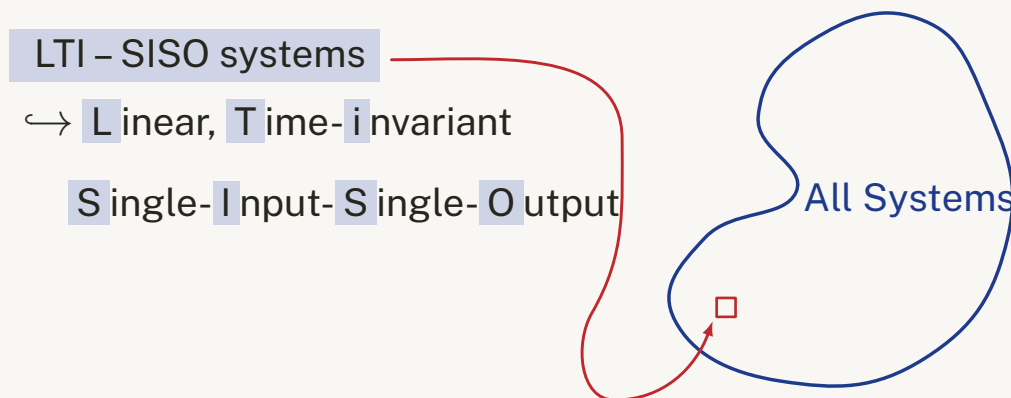
Statement	True	False
The system is linear	×	
The system is static		×
The system is time-invariant	×	

2 | Linearization

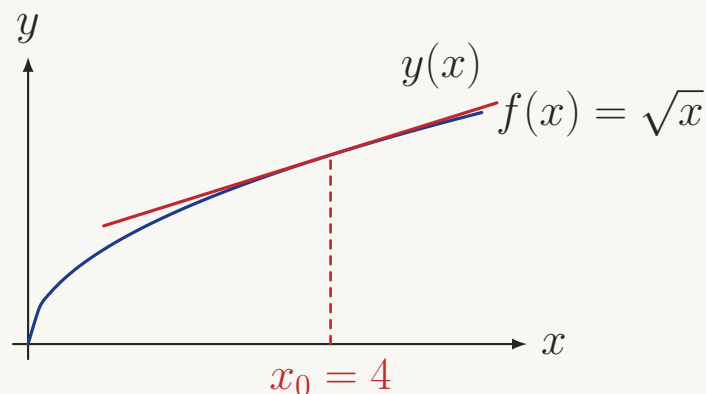
Motivation...

Real systems are rarely already linear — before we can use our SISO-LTI toolbox, we first need to linearize them around a point.

In this course (CS1), we will work with:



Linearization warm-up



→ We can approximate $f(x)$ at any point x_0 with a tangent function.

↪ Let's choose $x_0 = 4$.

General formula of a straight line: $y = mx + b$

$$m = \frac{df(x)}{dx} = \frac{1}{2\sqrt{x}} \quad b = f(x)$$

⇒ linearization at x_0 :

$$y(x) = \underbrace{\frac{1}{2\sqrt{x_0}}}_{\text{slope}} \underbrace{(x - x_0)}_{\text{shift by } x_0} + \underbrace{\sqrt{x_0}}_{\text{offset}}$$

Let's generalize this

General structure of our system:

$$\dot{x}(t) = f(x(t), u(t)) \quad (\text{I})$$

$$y(t) = h(x(t), u(t)) \quad (\text{II})$$

To linearize, we use the Taylor series:

$$T(f(x)) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x)}{n!} \Big|_{x=x_0} (x - x_0)^n$$

Apply to (I) & (II)

Write out the full first-order Taylor expansion of both equations.

$$\dot{x}(t) = f(x_0, u_0) + \frac{\partial f(x, u)}{\partial x} \Big|_{(x_0, u_0)} (x - x_0) + \frac{\partial f(x, u)}{\partial u} \Big|_{(x_0, u_0)} (u - u_0) + \text{h.o.t.}$$

$$y(t) = h(x_0, u_0) + \frac{\partial h(x, u)}{\partial x} \Big|_{(x_0, u_0)} (x - x_0) + \frac{\partial h(x, u)}{\partial u} \Big|_{(x_0, u_0)} (u - u_0) + \text{h.o.t.}$$

→ neglect the h.o.t.

→ choose the linearization point at an **equilibrium point**:

$$(x_0, u_0) = (x_e, u_e), \text{ where } \dot{x}(t) = f(x_e, u_e) = 0$$

... which leaves us with

$$\dot{x}(t) = \frac{\partial f(x, u)}{\partial x} \Big|_{(x_e, u_e)} (x - x_e) + \frac{\partial f(x, u)}{\partial u} \Big|_{(x_e, u_e)} (u - u_e)$$

$$y(t) = h(x_e, u_e) + \frac{\partial h(x, u)}{\partial x} \Big|_{(x_e, u_e)} (x - x_e) + \frac{\partial h(x, u)}{\partial u} \Big|_{(x_e, u_e)} (u - u_e)$$

→ do a substitution:

$$\Delta x = x - x_e \Rightarrow \Delta \dot{x} = \dot{x}$$

$$\Delta y = y - y_e = y - h(x_e, u_e)$$

$$\Delta \dot{x} = \frac{\partial f(x, u)}{\partial x} \Big|_{(x_e, u_e)} \Delta x + \frac{\partial f(x, u)}{\partial u} \Big|_{(x_e, u_e)} \Delta u$$

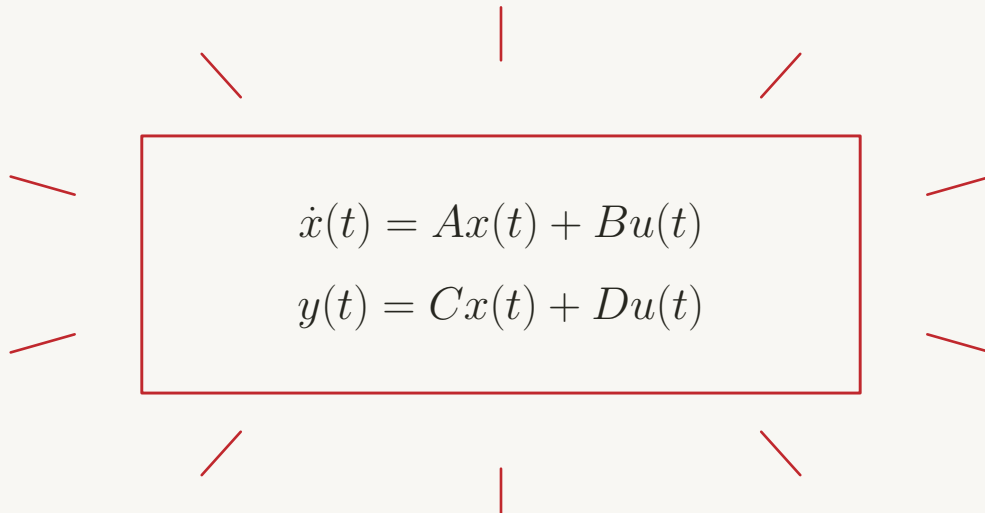
$$\Delta y = \frac{\partial h(x, u)}{\partial x} \Big|_{(x_e, u_e)} \Delta x + \frac{\partial h(x, u)}{\partial u} \Big|_{(x_e, u_e)} \Delta u$$

→ rename out of convention:

$$\Delta x = x, \quad \left. \frac{\partial f}{\partial x} \right|_{(x_e, u_e)} = A, \quad \left. \frac{\partial f}{\partial u} \right|_{(x_e, u_e)} = B$$

$$\Delta y = y, \quad \left. \frac{\partial h}{\partial x} \right|_{(x_e, u_e)} = C, \quad \left. \frac{\partial h}{\partial u} \right|_{(x_e, u_e)} = D$$

⇒ we arrive at the standard form for a linearized system:



$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t) \end{aligned}$$

$$x(0) = x_0$$

→ for the general case with multiple x_i & u_i , we must evaluate the **Jacobi matrices**:

$$A = \left. \frac{\partial f(x, u)}{\partial x} \right|_{(x_e, u_e)} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{pmatrix} \in \mathbb{R}^{n \times n}$$

$$B = \left. \frac{\partial f(x, u)}{\partial u} \right|_{(x_e, u_e)} = \begin{pmatrix} \frac{\partial f_1}{\partial u_1} & \dots & \frac{\partial f_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \dots & \frac{\partial f_n}{\partial u_m} \end{pmatrix} \in \mathbb{R}^{n \times m}$$

$$C = \left. \frac{\partial h(x, u)}{\partial x} \right|_{(x_e, u_e)} = \begin{pmatrix} \frac{\partial h_1}{\partial x_1} & \dots & \frac{\partial h_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial x_1} & \dots & \frac{\partial h_p}{\partial x_n} \end{pmatrix} \in \mathbb{R}^{p \times n}$$

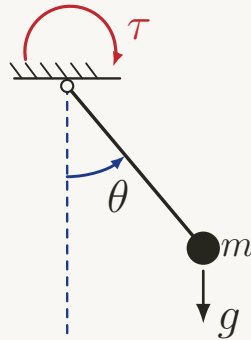
$$D = \left. \frac{\partial h(x, u)}{\partial u} \right|_{(x_e, u_e)} = \begin{pmatrix} \frac{\partial h_1}{\partial u_1} & \dots & \frac{\partial h_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial u_1} & \dots & \frac{\partial h_p}{\partial u_m} \end{pmatrix} \in \mathbb{R}^{p \times m}$$

3 | Equilibria

What we've used so far:

$$\dot{x} = f(x_e, u_e) = 0$$

EX. Pendulum



→ state: $x = (\dot{\theta}, \theta)$

→ input: τ

When to use linearization... operation point...

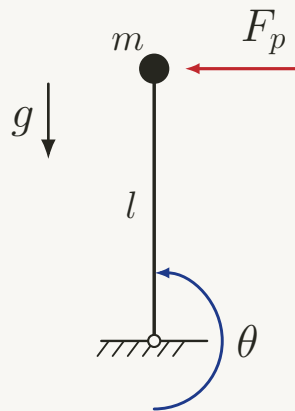
→ for which states can we have $\dot{\theta} = 0$? → $\forall \theta$

→ for which states, with zero input $\tau = 0$, does $\dot{\theta} = 0$ hold?

$x_{e1} = (0, 0)$ — **stable** $x_{e2} = (\pi, 0)$ — **unstable**

4 | Example

Pendulum Linearization:



$$\cdot u(t) = F_p(t)$$

$$\cdot x_1(t) = \theta(t)$$

$$\cdot x_2(t) = \dot{x}_1(t) = \dot{\theta}(t)$$

$$\cdot y(t) = \theta(t)$$

$$\rightarrow x(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} \Rightarrow \dot{x}(t) = \begin{pmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{pmatrix} = \begin{pmatrix} x_2(t) \\ -\frac{g}{l} \sin(x_1(t)) + \frac{1}{ml} \cos(x_1(t)) u(t) \end{pmatrix}$$

$$\rightarrow y(t) = x_1(t)$$

⇒ **Change of tactics!**

$$\dot{x}(t) = f(x(t), u(t))$$

$$y(t) = h(x(t), u(t))$$

⇒

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

1) Choose desired **operation point**: $x_e = \begin{pmatrix} \pi \\ 0 \end{pmatrix}$, $u_e = 0$

2) Evaluate Jacobi matrices:

$$A = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{pmatrix} \bigg|_{(\pi, 0, 0)} = \begin{pmatrix} 0 & 1 \\ g/l & 0 \end{pmatrix}$$

$$B = \begin{pmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \end{pmatrix} \bigg|_{(\pi, 0, 0)} = \begin{pmatrix} 0 \\ -1/(ml) \end{pmatrix}$$

$$C = \left(\frac{\partial h}{\partial x_1} \quad \frac{\partial h}{\partial x_2} \right) \bigg|_{(\pi, 0, 0)} = \begin{pmatrix} 1 & 0 \end{pmatrix}$$

$$D = \frac{\partial h}{\partial u} \bigg|_{(\pi, 0, 0)} = 0$$

⇒ the linearized pendulum:

$$\begin{aligned} \dot{x}(t) &= \begin{pmatrix} 0 & 1 \\ g/l & 0 \end{pmatrix} x(t) + \begin{pmatrix} 0 \\ -1/(ml) \end{pmatrix} u(t) \\ y(t) &= \begin{pmatrix} 1 & 0 \end{pmatrix} x(t) \end{aligned}$$

QUIZ

Q4) For a time-invariant system we can generalize the initial condition as follows: $x(t_0) = x_0 \Rightarrow x(0) = x_0$.

True / False

Q5) The following system can be strictly causal:

$$\dot{x}(t) = Ax + Bu, \quad y(t) = Cx + Du \quad \text{with } A, B, C, D \neq 0$$

True / False

Q6) These three terms mean the same: Linearization point / Equilibrium point / operation point.

True / False